

SPECTRUM: CORRECT-SOLUTION RETENTION IN LOOPED SELF-DISTILLATION

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ABSTRACT

A model that learns from its own outputs inherits more than their correctness: it inherits which solutions it produces. We study *Looped Self-Distillation*, in which a model repeatedly generates a synthetic corpus and learns from it, and identify a consequential separation: correctness can improve while the breadth of correct implementations contracts. We introduce SPECTRUM, a generation-time intervention that re-estimates loss-sensitive key/value geometry from a fixed reference anchor at each round and converts it into full-rank proximal spectral modulation. Every generated completion trains a single student, without rollout scoring or filtering; subsequent inference uses the native model. After five rounds of Qwen2.5-Coder-1.5B-Instruct on MBPP, SPECTRUM retains 89.9% of the initial model’s 64-sample correct AST richness, compared with 66.4% for plain self-distillation and 65.5% for a subspace-projection control. It is the only arm in this comparison whose pass@64 point estimate exceeds the initial model’s. The diversity advantage persists at matched correct-sample counts and is also observed on two transfer benchmarks, with dataset-dependent correctness trade-offs. These findings make correct-solution retention an explicit objective of repeated self-distillation and show that changing the generation geometry can improve what subsequent students retain.

1 INTRODUCTION

An improving model is also becoming a different teacher. When a language model generates its own training data, the distribution it learns at one round determines the examples available at the next. Repeating this process is attractive: the same model supplies both the generator and the student, and each update can become the starting point for another. Yet an increase in correctness need not make the resulting teacher a better source of diverse solutions. A model may become more reliable at producing a few familiar implementations while losing access to other correct ones.

This distinction matters beyond code. Mathematical problems can admit different proofs, planning tasks different valid plans, and scientific questions different explanatory paths. Such alternatives supply material for search, verification, and subsequent learning. We study coding because executable tests and program structure make the distinction measurable: *how often does a model produce a correct program, and how broadly does it sample within its correct programs?* Even the entire pass@ k curve cannot identify the latter distribution. Holding the total probability of correctness fixed leaves pass@ k unchanged when probability moves between correct implementations.

Our experiments expose this separation during repeated learning (Figure 1). After five rounds, plain self-distillation increases pass@1 from 37.94% to 41.45%, while correct AST richness at 64 samples falls from 12.802 to 8.506 classes per task. Thus the model improves on a familiar headline metric while the breadth of its correct outputs contracts by a third. This is especially consequential in a loop: the next training corpus is drawn from the changed distribution, not from the accuracy score.

Iterative self-distillation itself has precedents. Earlier work iterates the synthesis of contexts and rationales, improves code-data synthesizers, or refreshes a self-teacher during reasoning distillation (Rao et al., 2023; Zhang et al., 2025; He et al., 2026; Sang et al., 2026). Diversity loss has also been established for self-distillation with correct demonstrations supplied to a teacher (Nicolicioiu et al., 2026). We ask a more specific question: *when every raw completion is retained for training, can the generation mechanism preserve more correct-solution breadth through repeated student updates?*

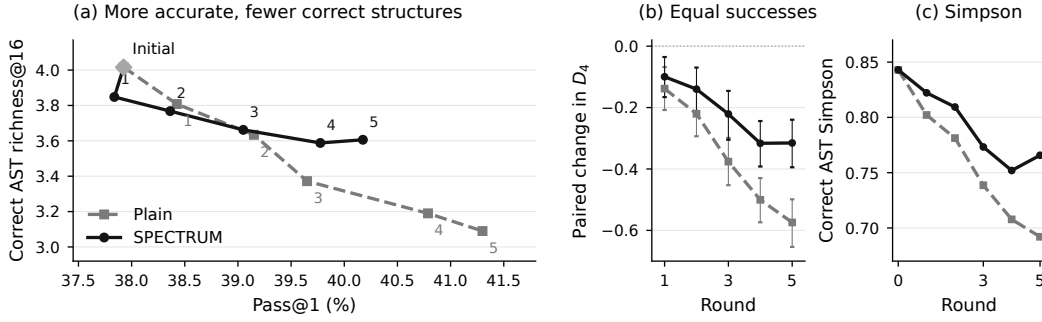


Figure 1: **Learning to succeed more often can narrow the successful outputs.** Five rounds of MBPP self-distillation, evaluated with 16 samples on each of 500 tasks. (a) The native students move toward higher pass@1 and lower correct AST richness; numbers denote rounds. (b) Changes in richness at four correct draws, paired against the initial model on the common eligible tasks of each contrast; bars are pointwise 95% task-bootstrap intervals. (c) Correct-output AST Simpson diversity also decreases (eligible tasks have at least two correct samples). Panels (a,c) show point estimates. One training seed; AST classes measure implementation structure.

This question separates a distributional failure from any particular rule for selecting successful rollouts.

We formulate this study as *Looped Self-Distillation*: an outer learning process that repeatedly constructs a generation policy, samples a corpus, and updates one native student. The loop reuses the learning procedure while changing its model state and synthetic data. Within this setting, SPECTRUM uses a small, fixed reference anchor to estimate the current model’s loss-sensitive key/value geometry. A continuous proximal gain modulates that geometry without assigning an exact zero gain to any local direction at finite strength. The model generates the next corpus under this temporary intervention, then learns from every raw completion through ordinary single-LoRA fine-tuning. Reference examples are reused throughout; no new annotations or quality judgments of the generated corpus are introduced between rounds.

The resulting student retains substantially more correct implementation breadth. At round five, SPECTRUM preserves 89.9% of initial 64-sample richness, improves pass@64 by 2.2 percentage points over Plain, and has a paired richness advantage when both methods are compared at the same number of correct draws. These benefits are measured after removing the generation intervention. Pass@1 is 1.12 points below Plain and 2.39 points above the initial model, making the result a measurable retention–correctness trade-off. Our contributions are:

1. **A longitudinal evaluation of solution retention.** We separate correctness probability from the distribution within correct outputs, and document their divergent evolution across five complete self-distillation rounds.
2. **A generation intervention for the loop.** We introduce SPECTRUM, combining a fixed reference anchor, per-round geometry estimation, and full-rank proximal spectral modulation before single-student learning.
3. **Evidence at three complementary endpoints.** We measure total-budget richness, correct-count-matched richness, and multi-sample success in native students, with a projection ablation, an SSD comparison, and frozen-student transfer to HumanEval+ and APPS Intro.

2 WHAT SHOULD A SELF-DISTILLATION LOOP RETAIN?

For a task x , let $v_x(y) \in \{0, 1\}$ denote passing its recorded verifier, and let $\phi(y)$ identify an implementation class. In coding, ϕ is a normalized AST fingerprint. Write $a_\theta(x) = \Pr_{\pi_\theta}[v_x(Y) = 1]$ and, when $a_\theta(x) > 0$,

$$q_{\theta,j}(x) = \Pr_{\pi_\theta}[\phi(Y) = j \mid v_x(Y) = 1], \quad \Pr[v_x(Y) = 1, \phi(Y) = j] = a_\theta(x)q_{\theta,j}(x). \quad (1)$$

The first factor is how frequently the model succeeds; the second is how it distributes its successes. We suppress x below and assume independent draws from a fixed decoding policy.

Proposition 1 (Correctness does not identify correct-solution breadth). *For fixed $a > 0$, $\text{pass}@k = 1 - (1 - a)^k$ is independent of q . Expected correct-class counts under k total draws and b correct draws are, respectively,*

$$C_k(a, q) = \sum_j [1 - (1 - aq_j)^k], \quad D_b(q) = \sum_j [1 - (1 - q_j)^b]. \quad (2)$$

Distributions with identical $\text{pass}@k$ for every k can have different D_b for any $b \geq 2$.

For example, at $a = 0.4$, concentrating all correct mass on one class gives $D_4 = 1$, whereas distributing it uniformly across four classes gives $D_4 = 2.734375$. Both policies have $\text{pass}@4$ equal to 0.8704. These are elementary occupancy identities, not a model-specific assumption. They explain why correctness alone cannot diagnose contraction within correct outputs.

We use descriptive names: *correct AST richness@k* for C_k and *correct-count-matched AST richness@b* for D_b . Standard rarefaction estimators compute them from finite sample pools (Hurlbert, 1971); definitions and proofs appear in Appendix A. C_k answers what breadth a fixed generation budget buys. D_b asks whether a difference remains after equalizing successful draws. Conditional comparisons use tasks eligible for both methods and report their number. Neither metric identifies semantic algorithm equivalence.

For a sequence of students $\theta_0, \dots, \theta_T$, the *retention profile* records $(\text{pass}@k, C_k, D_b)$ under a common evaluation protocol. Relative richness retention is $R_{t,k} = C_k(\theta_t)/C_k(\theta_0)$ when defined. This is retention of expected class *count*, not identity overlap with the initial classes. It measures a finite-budget repertoire rather than the total support of the model’s distribution.

3 LOOPED SELF-DISTILLATION WITH A FIXED REFERENCE ANCHOR

Let $\mathcal{Q}$ be a fixed set of synthesis prompts and θ_t the current native student. A generation constructor $\mathcal{G}$ and a learner $\mathcal{L}$ define one round:

$$\tilde{\pi}_t = \mathcal{G}(\theta_t; \mathcal{A}), \quad \mathcal{D}_t = \{(x, Y_{x,s}) : x \in \mathcal{Q}, Y_{x,s} \sim \tilde{\pi}_t\}_{s=1}^S, \quad \theta_{t+1} = \mathcal{L}(\theta_t, \mathcal{D}_t). \quad (3)$$

Here $\mathcal{A} = \{(x_i, y_i^*)\}$ is optional fixed calibration information. Plain and decoding-based self-distillation ignore $\mathcal{A}$; SPECTRUM uses reference completions to shape generation. In our protocol the learner retains every generated record, warm-starts from θ_t , and updates a single LoRA adapter (Hu et al., 2022). The merged student becomes the next generator’s starting point. Historical corpora are not accumulated.

Why loop? A one-round evaluation measures a teacher–student transition. A loop additionally asks what happens when the resulting student becomes the teacher again. It exposes whether an immediate gain provides a useful distribution for subsequent learning. This is an operational setting for studying repeated self-distillation, building on iterative self-training; monotonic self-improvement is an empirical question, not part of its definition. The term also distinguishes an outer *learning* loop, with changing parameters, from recurrent-depth transformers that repeat computation within inference (Dehghani et al., 2019).

Where supervision enters. SPECTRUM repeatedly uses the *same* reference anchor to recalibrate the evolving model. The anchor is external supervision fixed before the loop; the geometry derived from it changes each round. Generated completions are never scored, ranked, filtered, or reweighted by a verifier, judge, or strategy label. Test outcomes are used only for evaluation. This distinguishes fixed calibration guidance from ongoing assessment of synthetic samples. It does not make the method fully unsupervised. The student objective is sequence-level cross-entropy on self-generated data, rather than reward optimization or a demonstration-conditioned teacher’s token-level KL loss.

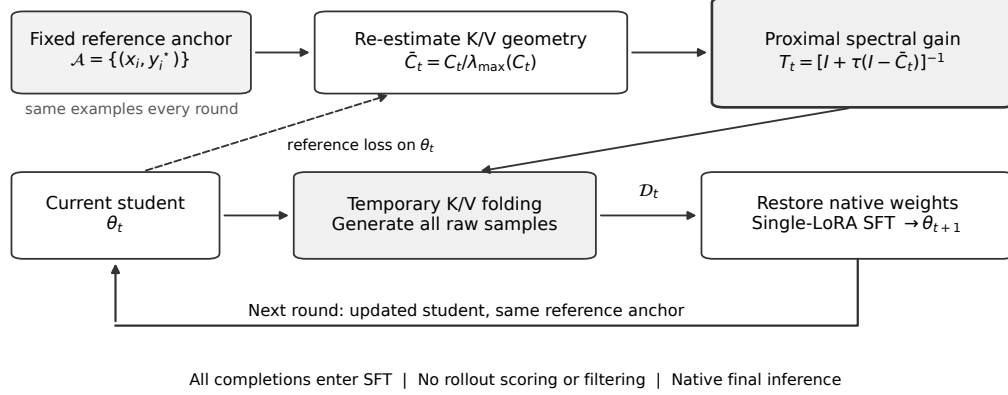
SPECTRUM inside Looped Self-Distillation

Figure 2: **Fixed information, evolving geometry, one student per round.** The reference anchor remains unchanged while the current student’s completion loss yields a new K/V second moment. SPECTRUM converts that geometry into a proximal gain, folds it into generation weights temporarily, and samples a raw corpus. Native weights are restored before LoRA learning. The updated student supplies the next round and is evaluated through ordinary inference. Dashed input denotes gradient-based calibration; the bottom return arrow denotes a parameter update. No sample-selection stage lies on the learning path.

4 SPECTRUM: PROXIMAL SPECTRAL MODULATION

4.1 REFERENCE GEOMETRY WITHOUT ASSIGNING SOLUTION STRATEGIES

For anchor example (x_i, y_i^*) , let $\ell_i(\theta_t)$ be mean negative log likelihood over its completion targets. Prompt and padding targets are masked. For selected K/V module ℓ , collect $g_{iu}^{(\ell)}$, the gradient of this scalar loss with respect to the module’s native linear output at every nonpadding position u . Following reference-loss-gradient calibration of K/V directions (Hao et al., 2026), compute

$$C_{t,\ell} = \frac{1}{M_\ell} \sum_{i,u \text{ nonpad}} g_{iu}^{(\ell)} g_{iu}^{(\ell)\top}, \quad \bar{C}_{t,\ell} = C_{t,\ell} / \lambda_{\max}(C_{t,\ell}). \quad (4)$$

M_ℓ counts the included positions, including those with zero gradient. Collection includes prompt positions: masking their loss targets does not remove their influence on completion loss. The statistic is an uncentered gradient second moment, with normalized spectrum in $[0, 1]$.

High sensitivity indicates that the reference loss responds strongly to a direction. Low sensitivity on a finite anchor does not establish that the direction is useless for other valid implementations. We therefore use the spectrum continuously. There is no assignment of ten solutions to ten subspaces, no strategy annotation, and no adapter pool.

For a unit direction v , the quadratic form $v^\top C_{t,\ell} v$ is the mean squared directional derivative $(g_{iu}^{(\ell)\top} v)^2$. It measures sensitivity, not a reward or a signed improvement direction. Normalization removes the overall gradient scale and makes τ control relative attenuation. Re-estimating this statistic matters because the student parameters, activations, and reference losses change after learning. Reusing the anchor holds its information budget fixed while measuring sensitivity in the current model; fixing the operator instead would reuse geometry estimated under earlier parameters.

4.2 A FULL-RANK PROXIMAL GAIN

For row activation $h \in \mathbb{R}^{1 \times d}$, define the modulated state by

$$\arg \min_z \left\{ \frac{1}{2} \|z - h\|_2^2 + \frac{\tau}{2} z(\mathbf{I} - \bar{C})z^\top \right\} = hT_\tau, \quad T_\tau = [\mathbf{I} + \tau(\mathbf{I} - \bar{C})]^{-1}, \quad \tau \geq 0. \quad (5)$$

This is a classical quadratic proximal map (Parikh & Boyd, 2014), applied here to the generation geometry of a self-distillation loop. The first term limits departure from the current state; the second attenuates directions less emphasized by the reference statistic. If $\overline{C} = U \text{diag}(\mu_j) U^\top$, its gain is

$$T_\tau = U \text{diag}\left(\frac{1}{1 + \tau(1 - \mu_j)}\right) U^\top. \quad (6)$$

The maximal-sensitivity directions retain unit gain, while every other direction retains a positive response at finite τ . Modulation strength controls this departure continuously; it is not a prescribed count of solution modes.

For illustration, normalized eigenvalues $(1, \frac{1}{2}, 0)$ at $\tau = 1$ produce gains $(1, \frac{2}{3}, \frac{1}{2})$. The least-sensitive component is attenuated but remains available to downstream computation. This treatment is useful when an anchor covers only some valid implementations: absence of measured sensitivity need not become an irreversible local deletion. It supplies a design rationale for retention, rather than assuming that eigenvectors correspond to named algorithms.

Proposition 2 (Local distinguishability and calibration stability). *For $\overline{C} \succeq 0$ with eigenvalues in $[0, 1]$ and finite $\tau \geq 0$, Equation (5) has a unique solution. Its gains lie in $[1/(1 + \tau), 1]$, and*

$$\frac{\|\delta\|_2}{1 + \tau} \leq \|\delta T_\tau\|_2 \leq \|\delta\|_2, \quad \|T_\tau - \mathbf{I}\|_2 \leq \frac{\tau}{1 + \tau}. \quad (7)$$

For normalized PSD statistics A, B with spectra in $[0, 1]$, $\|T_\tau(A) - T_\tau(B)\|_2 \leq \tau\|A - B\|_2$.

The operator thus avoids exact local erasure and is continuous in its normalized geometry without an eigengap assumption. These are properties of a local linear map; preservation of semantic solution classes through attention, decoding, and training remains an empirical question. Appendix B proves the bounds and states their scope.

4.3 FROM A TEMPORARY GENERATOR TO A NATIVE STUDENT

For $h = xW^\top + b^\top$, fold the row operator into $W' = T_\tau^\top W$ and $b' = T_\tau^\top b$. Then $xW'^\top + b'^\top = hT_\tau$. Folding acts before rotary position transformations and key normalization, and introduces no extra activation hook. After generation, restore the original weights and train on all raw completions, including unsuccessful, empty, and duplicate records.

Let $z = (x, y)$ be a tokenized training record, $\mathcal{T}(z)$ its nonpadding causal targets, and Δ the effective LoRA weight update. The learner optimizes

$$\mathcal{L}_t(\Delta) = -\frac{1}{|\mathcal{D}_t|} \sum_{z \in \mathcal{D}_t} \frac{1}{|\mathcal{T}(z)|} \sum_{u \in \mathcal{T}(z)} \log p_{\theta_t + \Delta}(z_u \mid z_{<u}). \quad (8)$$

Our recorded SFT loss includes prompt targets, whereas calibration masks them. Every record has the same example weight; the spectral gain changes the sampled corpus, not a sample’s loss coefficient. Thus frequently generated implementations contribute through their empirical frequency, with no routing of solutions to separate adapters. After the fixed training schedule, merging Δ_t yields θ_{t+1} . Its geometry is recalibrated from the same $\mathcal{A}$ next round. Native evaluation uses neither the gain nor the anchor, so the measured endpoint is the distribution learned by the student.

The K and V interventions play complementary roles: changing keys alters query–key compatibility, while changing values alters the information aggregated by attention. Their effect depends on the input and current hidden state, beyond a global temperature applied to output logits. Across rounds, only the learned student is carried forward. Restoring native weights prevents direct composition of successive spectral gains; the next operator is estimated afresh from the updated student. This construction makes the synthetic corpus the channel through which a temporary geometric intervention can persist in later behavior.

5 EXPERIMENTS

5.1 PROTOCOL AND RESEARCH QUESTIONS

The main experiment uses Qwen2.5-Coder-1.5B-Instruct (Hui et al., 2024) and MBPP’s full test split of 500 tasks (Austin et al., 2021). After removing three exact normalized train/test prompt overlaps,

Table 1: **Final native students after five MBPP rounds.** All-task means from 64 draws on each of 500 tasks; higher is better. D_4 means use each row’s eligible population, whose size is reported. Paired conditional effects appear in Figure 3 and Appendix D. Bold identifies the better trained student in each performance row, without asserting statistical dominance.

Metric	Initial model	Plain	SPECTRUM
pass@1 (%)	37.94	41.45	40.33
pass@8 (%)	61.10	61.86	61.89
pass@16 (%)	65.69	65.64	66.23
pass@64 (%)	72.00	70.40	72.60
Correct AST richness@64 (C_{64})	12.802	8.506	11.510
Relative richness retention (%)	100.0	66.4	89.9
Correct-count-matched richness@4 (D_4)	3.358	2.856	3.143
D_4 eligible tasks	322	321	323
Solved tasks / 500	360	352	363

the training pool contains 291 synthesis tasks, 50 fixed calibration tasks, and 30 validation tasks. These sets are disjoint. Each of five rounds generates 16 raw completions per synthesis task and runs one LoRA epoch: 4,656 records per round, 23,280 in total. The completed setting uses $\tau = 1$, middle and last decoder-layer K/V modules, and LoRA rank 8. It does not choose τ anew from test outcomes.

All native students use common evaluation decoding: temperature 0.8, top- p 0.95, and no top- k truncation. The historical trajectory has 16 samples per task per round; a separate final supplement evaluates the initial and round-five checkpoints with 64 samples per task. These pools are kept distinct. The main training seed is 43; confidence intervals resample evaluation tasks, not training seeds. We report available results rather than treating configured runs as completed evidence. Appendix C gives the exact losses, prompts, and budgets.

We ask four questions: **RQ1**, does spectral generation improve breadth retained by a native student? **RQ2**, does the benefit survive matching correct-sample counts and changing sample budget? **RQ3**, what trade-offs arise under a projection control and decoding-based SSD? **RQ4**, does the learned effect transfer beyond the training benchmark?

5.2 RQ1: CAN THE LEARNED STUDENT RETAIN MORE CORRECT-SOLUTION BREADTH?

Table 1 gives the final 64-sample comparison. SPECTRUM retains 11.510 correct AST classes per task, versus 8.506 for Plain: a 35.3% relative increase, or a paired difference of 3.004 classes with 95% task-bootstrap interval [2.568, 3.494]. Relative to the initial model, this is 89.9% versus 66.4% richness retention. The result is partial preservation: SPECTRUM still loses 10.1% of the initial richness.

The accuracy profile explains what this retention buys. SPECTRUM solves 363 tasks at 64 samples, compared with 352 for Plain, a pass@64 gain of 2.2 points [0.4, 4.2]. Its point estimate also exceeds the initial model’s 360 tasks, although that difference is inconclusive (+0.6 points, [−1.8, 3.0]). Pass@1 improves by 2.39 points over initialization but is 1.12 points below Plain ([−1.61, −0.63] after rounding). The recorded one-point noninferiority criterion against Plain is therefore not met. The supported gain is more retained breadth and stronger high-budget success with a small single-sample accuracy cost.

The 16-sample trajectory connects this endpoint to repeated learning. Plain’s correct AST richness declines from 4.016 to 3.090, while SPECTRUM ends at 3.606. SPECTRUM has higher richness than Plain at every recorded post-training round. The corresponding relative retentions are 76.9% and 89.8% at this smaller budget. Figure 1 also shows less contraction in matched-correct richness and less concentration in the successful outputs. All these measurements concern native students; the generation intervention is absent during their evaluation.

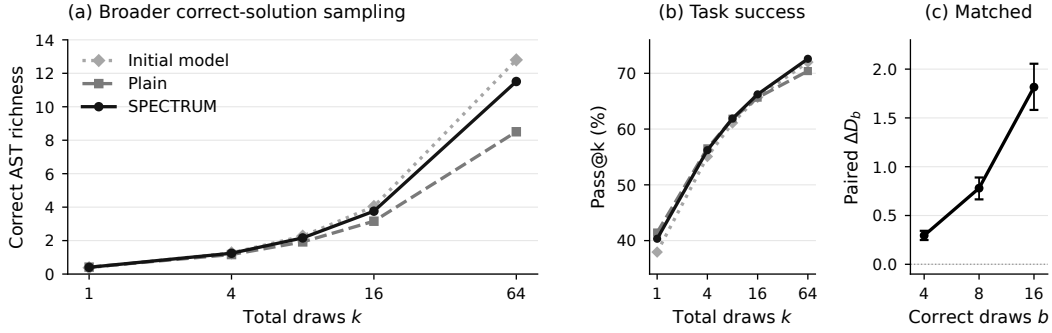


Figure 3: **Breadth becomes consequential as sampling budget grows.** (a,b) All-task means from the same round-five 64-sample pools at $k \in \{1, 4, 8, 16, 64\}$; the horizontal axes are logarithmic and connecting lines only join recorded estimates. (c) Paired SPECTRUM-minus-Plain richness gains after fixing the number of correct draws; bars are 95% task-bootstrap intervals on 316, 296, and 254 common eligible tasks for $b = 4, 8, 16$. No additional outputs were generated for these analyses.

Table 2: **Generation choices under the completed 16-sample protocol.** Round-five native students on 500 MBPP tasks. SSD is a decoding-recipe adaptation run separately with the same task and training budgets; its initial resampling is reported in Appendix D. The projection row is a design control. All rows use the common post-training decoder. D_4 is a marginal eligible-task mean.

Generation design	pass@1 (%)	pass@16 (%)	C_{16}	D_4	Eligible
Plain	41.30	66.00	3.090	2.747	268
SSD recipe	35.96	65.60	3.710	3.268	255
SPECTRUM	40.18	67.20	3.606	2.997	263
Subspace-projection control	41.35	65.60	3.066	2.690	262

5.3 RQ2: IS THE GAIN EXPLAINED BY MORE SUCCESSFUL SAMPLES?

Correct-sample matching isolates variety within successful outputs. At four correct draws, SPECTRUM exceeds Plain by 0.295 expected AST classes, with interval $[0.249, 0.343]$, on 316 common eligible tasks. At eight and sixteen correct draws, the paired gains are 0.780 and 1.816, on 296 and 254 tasks (Figure 3c). Thus the gain in C_{64} cannot be explained solely by obtaining more correct draws. Eligibility changes with b ; these conditional estimates do not describe all 500 tasks.

The budget curves reveal complementary behavior. At $k = 1$, C_1 is correctness itself and favors Plain. At every displayed larger budget, SPECTRUM yields more distinct correct structures; its pass@ k advantage is clearest at the largest measured budget. Matched-correct gains also grow with b , showing that the difference extends beyond a single extra rare structure. These findings support using success and conditional richness together; neither alone fully describes a generator’s value for the next loop.

5.4 RQ3: WHAT DOES THE GENERATION DESIGN CHANGE?

We replace the proximal gain by a top-half rank projector as a direct design control, using the same reference geometry and learning schedule. This control follows the subspace-projection construction of Hao et al. (2026); it is a declared reconstruction with our completion-loss and insertion-point choices, not an official reproduction. Its round-five C_{64} is 8.388 (65.5% retention), versus 11.510 for SPECTRUM. The soft gain improves richness by 3.122 classes and pass@64 by 2.4 points while lowering pass@1 by 1.43 points. This contrast supports retaining a continuous local response; it does not isolate spectral orientation from attenuation strength.

For a decoding-based comparator, we loop SSD’s temperature/truncation recipe (Zhang et al., 2026): training generation uses temperature 1.5, top- p 0.8, and top- k 20; evaluation returns to the common

Table 3: **Frozen-student transfer after MBPP learning.** Sixteen samples per task; no transfer training or recalibration. C_{16} and success average over all tasks. D_4 is the eligible-task marginal mean; common-task contrasts are given in the text. Higher is better, except the eligibility count is descriptive.

Benchmark	Student	pass@1 (%)	pass@16 (%)	C_{16}	D_4	Eligible
HumanEval+ (164)	Initial	53.09	85.37	5.768	3.410	116
	Plain	53.47	86.59	5.848	3.422	120
	SPECTRUM	51.94	86.59	6.000	3.497	114
APPS Intro (200)	Initial	15.47	41.50	2.070	3.772	50
	Plain	15.19	35.50	1.930	3.632	46
	SPECTRUM	13.97	34.00	1.845	3.721	43

decoder. Table 2 shows a different trade-off. SSD retains more richness than SPECTRUM at 16 draws, but its pass@1 is 4.21 points lower. SPECTRUM exceeds Plain in richness while retaining more accuracy than this SSD recipe. The study therefore establishes distinct operating points, not universal dominance over diversity-oriented decoding. The completed runs do not include a strength sweep or controls with randomized directions, so necessity of the calibrated orientation is unresolved.

5.5 RQ4: DOES CORRECT-SOLUTION RETENTION TRANSFER?

We evaluate the frozen MBPP-trained round-five students on all 164 HumanEval+ tasks and a 200-task APPS Intro subset, with 16 outputs per task and no further learning. HumanEval+ uses EvalPlus 0.3.1 and its enhanced tests (Liu et al., 2023). APPS uses the recorded adapter to a Python function interface and is not an official APPS leaderboard protocol (Hendrycks et al., 2021). Table 3 separates the two settings.

On HumanEval+, SPECTRUM matches Plain’s pass@16 and has a paired D_4 advantage of 0.080 classes [0.021, 0.142] on 111 tasks. On APPS Intro, the paired advantage is 0.104 [0.007, 0.205] on 42 tasks, but all-task success and C_{16} are lower. This contrast is informative: richer successful outputs do not automatically produce broader task coverage under distribution shift. It supports transfer of the conditional retention effect while locating a boundary on task-level gains. These intervals are post-hoc task bootstraps from saved per-task records. Appendix E adds equal-correct-count examples selected by a deterministic rule, with both directions of the effect shown.

6 RELATED WORK

Iterative self-training and self-distillation. STaR and ReST-EM establish repeated generation and learning as a route to improving reasoning from self-produced solutions (Zelikman et al., 2022; Singh et al., 2024). Iteration also exists explicitly within self-distillation: Rao et al. (2023) improve contexts and rationales using critics and filtering, while SCoder develops a code-data synthesizer through multiple checkpoints, scoring, and sample selection (Zhang et al., 2025). These works motivate studying the student as a future data source. Our loop exposes the evolution of that source under a fixed prompt and reference budget, measuring which correct implementation breadth survives each complete corpus-generation/student-update cycle. It concerns outer parameter learning, whereas recurrent-depth transformers repeat computation within a forward process (Dehghani et al., 2019).

The source of self-distillation feedback. Self-generated targets can support different learning objectives. SD-Zero turns reward-conditioned self-revision into dense supervision and synchronizes teachers during self-evolution (He et al., 2026). CRISP periodically refreshes an instruction-conditioned self-teacher and uses on-policy token-level KL training to compress reasoning, without correctness filtering (Sang et al., 2026). SSD trains on raw code samples whose distribution is shaped by temperature and truncation, and analyzes the tension between precision and exploration (Zhang et al., 2026). SPECTRUM shares sequence-level learning from raw outputs with SSD, while deriving its generation intervention from fixed reference-loss geometry. Consequently, its supervision budget and mechanism differ from both purely unlabeled self-training and demonstration-conditioned teacher

feedback. Avoiding rollout selection and refreshing a teacher are existing ingredients; the retention question concerns their repeated downstream effect.

Diversity under recursive learning. Nicolicioiu et al. (2026) identify reduced functional and semantic diversity in self-distillation conditioned on sampled correct demonstrations. Their analysis connects the optimal-policy bias to compatibility between a rollout and its demonstration. Our empirical loop omits this feedback mechanism and evaluates structural richness within correct outputs after repeated raw-output learning. UA-RL takes another route by rewarding rare solution strategies (Hu et al., 2026); SPECTRUM does not score or label generated strategies. More broadly, recursive model-collapse studies show distributional degradation under synthetic training, and retaining accumulated data can mitigate it (Shumailov et al., 2024; Gerstgrasser et al., 2024). Here the endpoint is conditional on correctness: implementation breadth can decline while accuracy improves. Jointly reporting $\text{pass}@k$, C_k , and D_b distinguishes successful-draw frequency from breadth within those draws, while keeping the historical-data policy fixed.

Spectral control and retained student behavior. Spectral activation editing uses internal geometry to control model outputs (Qiu et al., 2024). Capability-selective self-policy distillation constructs reference-gradient K/V subspaces for generation (Hao et al., 2026). We build on this calibration principle and apply a full-rank proximal response, with per-round geometry estimation and subsequent learning in one native student. Quadratic proximal maps and their closed forms are classical (Parikh & Boyd, 2014). Their role here is to regulate sensitivity-based attenuation without setting a local direction’s gain to zero. The operator bounds motivate the design; longitudinal, correct-count-matched student measurements establish the observed retention effect. This separation distinguishes a mathematically controlled intervention from an assumed semantic-diversity guarantee.

7 LIMITATIONS

The completed study uses one 1.5B backbone and one training seed, with two frozen-student transfer benchmarks. Task-bootstrap intervals characterize evaluation-task uncertainty rather than training variability. AST fingerprints measure implementation structure, not audited algorithm identities; passing finite tests is an operational correctness criterion. The findings therefore concern measured code repertoires, with extension to proofs or plans left open.

The fixed reference anchor supplies external information. Plain does not isolate its contribution from the operator; the projection control shares the anchor, but directional and strength-matched controls are absent. Equal record counts also give unequal token budgets: SPECTRUM generates 9.9% more training tokens than Plain (Appendix F). Full rank is a local property, and neither global diversity preservation nor monotonic improvement follows from it. The lower $\text{pass}@1$ and APPS task coverage identify boundaries of the observed retention benefit.

8 CONCLUSION AND FUTURE WORK

Looped Self-Distillation makes each student the source of its next training distribution. Our study shows that this process can improve correctness while contracting correct implementation breadth. SPECTRUM combines a fixed reference anchor with evolving loss-sensitive geometry and a full-rank proximal gain. After five rounds it retains 89.9% of initial 64-sample richness, compared with 66.4% for Plain; matched-correct analyses locate a benefit inside successful outputs, and native inference shows that it survives student learning.

Future work can connect these endpoints more directly by measuring immediate generator changes and their transmission into successive students. Matched-token studies, fixed-geometry controls, and alternative spectral orientations would isolate the contribution of recalibration and directional structure. Extending the loop across model families and to verifiable proofs or plans would test whether retaining multiple valid solutions is a general design principle for repeated self-improvement.

REPRODUCIBILITY STATEMENT

The appendix specifies the metric estimators, proofs, information flow, losses, data splits, prompts, decoding, and resource accounting. The accompanying analysis package records source hashes and regenerates every figure from saved results, including the transfer task bootstraps. This manuscript introduces no new training, generation, or program-verification runs.

AI USE STATEMENT

AI assistants supported literature retrieval, experiment-software development, mathematical exposition, manuscript drafting, and figure generation code. The paper’s empirical values come from the recorded experiments and explicitly identified analyses of saved results; no missing benchmark values were generated or inferred.

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A METRICS, ESTIMATORS, AND PROBABILITY ANALYSIS

A.1 CORRECTNESS AND CONDITIONAL IMPLEMENTATION DISTRIBUTIONS

For a fixed task, let $V = v_x(Y)$ and $J = \phi(Y)$. Equation (1) is simply $\Pr(V = 1, J = j) = \Pr(V = 1) \Pr(J = j \mid V = 1)$. If $a = 0$, unconditional correct richness is zero and q is undefined. We do not assign a fabricated conditional distribution to a task with no observed successes. For countably many classes the occupancy sums converge, because $1 - (1 - p)^k \leq kp$ gives $C_k \leq ka$ and $D_b \leq b$. These quantities count expected observed classes at the stated budget, not all possible implementations.

Proof of Proposition 1. Let I_j indicate observing at least one correct output from class j in k independent draws. Its expectation is $1 - (1 - aq_j)^k$. Linearity of expectation gives C_k . The same argument for b draws from the conditional correct distribution yields D_b . All k draws fail with probability $(1 - a)^k$, so $\text{pass}@k$ depends only on a . For $q = (1, 0, \dots)$, $D_b = 1$. For uniform mass on $J \geq 2$ classes, $D_b = J[1 - (1 - 1/J)^b] > 1$ when $b \geq 2$. Hence identical correctness at every budget does not identify conditional breadth. $\square$

More generally, spreading mass from a larger to a smaller conditional class, without reversing their order, cannot decrease D_b : $f(q) = 1 - (1 - q)^b$ is concave for $b \geq 2$. The same applies to C_k at fixed a and $k \geq 2$. This describes a change in a probability distribution; it does not assert that the spectral operator induces that change in every network.

A.2 FINITE-SAMPLE ESTIMATORS

Suppose a task has n generated outputs, m of which pass its verifier, and correct-class counts m_j with $\sum_j m_j = m$. We compute

$$\widehat{\text{pass}@k} = 1 - \frac{\binom{n-m}{k}}{\binom{n}{k}}, \quad 1 \leq k \leq n, \quad (9)$$

$$\widehat{C}_k = \sum_j \left[1 - \frac{\binom{n-m_j}{k}}{\binom{n}{k}} \right], \quad 1 \leq k \leq n, \quad (10)$$

$$\widehat{D}_b = \sum_j \left[1 - \frac{\binom{m-m_j}{b}}{\binom{m}{b}} \right], \quad 1 \leq b \leq m. \quad (11)$$

We set $\binom{u}{v} = 0$ for nonnegative integers $u < v$. Equation (9) is the standard $\text{pass}@k$ estimator (Chen et al., 2021). Equations (10)–(11) are rarefaction estimators (Hurlbert, 1971). A uniformly chosen subset of k outputs misses correct class j with probability $\binom{n-m_j}{k} / \binom{n}{k}$. Summing its complement gives the exact expected subset richness conditional on the recorded pool; under independent sampling it estimates the corresponding population quantity. Conditioning on correct outputs gives the analogous calculation for D_b .

At $k = n$, $\widehat{C}_n$ is the number of observed correct classes. At $k = 1$, $\widehat{C}_1 = m/n = \widehat{\text{pass}@1}$. At $b = 1$, $D_1 = 1$ wherever defined. Thus neither C_1 nor D_1 distinguishes how probability is distributed among multiple correct classes. The plotted budgets are those present in the archived estimates; we do not reconstruct a missing $\text{pass}@32$ value by interpolation.

A.3 AST FINGERPRINTS AND CONCENTRATION

Python programs are extracted using the recorded first-fenced-block rule. The primary fingerprint parses the extracted program and conservatively normalizes its AST. Formatting and docstrings are removed; local identifiers are normalized only when scope handling is unambiguous. Function names, external identifiers, literals, and uncertain scopes are retained. Consequently, distinct fingerprints can reflect structural implementation variants within one algorithm. Conversely, a fingerprint does not establish functional equivalence beyond the recorded tests. Parse failures do not silently become new correct classes. Secondary exact-program and control-flow fingerprints provide different granularities.

For Figure 1c, the saved correct-output Simpson statistic is

$$\hat{S} = 1 - \frac{\sum_j m_j(m_j - 1)}{m(m - 1)}, \quad m \geq 2. \quad (12)$$

It estimates the probability that two independently drawn correct outputs fall in different classes. Its aggregate uses tasks with enough correct samples and complete labels. It complements richness by emphasizing concentration, while sharing the limitations of the class definition. Exact-program, AST, and control-flow metrics need not move equally; the coarser control-flow comparisons do not establish a uniform increase in algorithmic families.

A.4 TASK AGGREGATION AND UNCERTAINTY

Pass@ k and C_k are macro-averaged over the full task set. A marginal D_b mean uses $\mathcal{E}_b(M) = \{x : m_x(M) \geq b\}$ for model M and reports its size. A paired contrast between M_1, M_2 averages task-level differences on $\mathcal{E}_b(M_1) \cap \mathcal{E}_b(M_2)$. Subtracting marginal means with different eligible sets is not this paired effect. Figure 1b pairs each round with initialization; its eligible population can change between rounds. It is a pointwise trajectory, not a single fixed cohort.

Archived MBPP intervals use 2,000 task-bootstrap resamples. For the new analysis of saved transfer metrics, we fix bootstrap seed 20260924 and also use 2,000 resamples, pairing by exact task ID and conditioning on common eligibility. Percentile endpoints are the 2.5th and 97.5th percentiles. The analysis script records the means, intervals, populations, and input hashes. These intervals describe evaluation-task uncertainty for the saved checkpoints. They do not measure independent training-seed variation, prove generality across architectures, or treat rounds as independent replications. They are pointwise exploratory intervals without multiplicity adjustment.

B OPERATOR DERIVATION AND MATHEMATICAL SCOPE

B.1 THE CALIBRATION LOSS

Let $\mathcal{T}_i$ contain the nonpadding completion targets of reference i . The scalar differentiated is

$$\ell_i(\theta) = -\frac{1}{|\mathcal{T}_i|} \sum_{u \in \mathcal{T}_i} \log \pi_\theta(y_{iu}^* \mid x_i, y_{i, < u}^*). \quad (13)$$

It is computed per example before collecting gradients. Every nonpadding K/V-output position contributes to Equation (4), including prompt positions and zero-gradient positions. Thus the loss mask and gradient-position mask are distinct. A large eigenvalue of C represents mean squared directional sensitivity. Opposite signs both contribute positively, so this statistic does not establish signed update agreement or correctness of a generated sample. Empty or zero-signal calibration is rejected; $C/\lambda_{\max}(C)$ otherwise would be undefined.

B.2 PROOF OF PROPOSITION 2

The Hessian of Equation (5) is $H = \mathbf{I} + \tau(\mathbf{I} - \overline{C})$, with eigenvalues $1 + \tau(1 - \mu_j) \in [1, 1 + \tau]$. It is positive definite, so the objective has a unique minimizer. Its first-order condition is $zH = h$, giving $z = hH^{-1}$. Diagonalizing H yields Equation (6). Since T_τ is symmetric positive definite, its singular values equal its gains, proving the two-sided norm bound. Also,

$$\|T_\tau - \mathbf{I}\|_2 = \max_j \frac{\tau(1 - \mu_j)}{1 + \tau(1 - \mu_j)} \leq \frac{\tau}{1 + \tau}. \quad (14)$$

The condition number is at most $1 + \tau$, and $T_0 = \mathbf{I}$. For normalized statistics A, B , the resolvent identity gives

$$T_\tau(A) - T_\tau(B) = \tau T_\tau(A)(A - B)T_\tau(B). \quad (15)$$

Both outer factors have operator norm at most one, which proves the stability bound without an eigenvector or eigengap assumption. $\square$

This last result applies to normalized statistics. If raw PSD matrices C, D both satisfy $\lambda_{\max} \geq \gamma > 0$, Weyl’s inequality and a triangle inequality additionally give

$$\left\| \frac{C}{\lambda_{\max}(C)} - \frac{D}{\lambda_{\max}(D)} \right\|_2 \leq \frac{2\|C - D\|_2}{\gamma}. \quad (16)$$

Combining the bounds yields a local perturbation bound $2\tau\|C - D\|_2/\gamma$. It is not a sampling-concentration theorem for the calibration estimator; it explicitly requires a nonvanishing normalization scale.

B.3 FOLDING, FINITE PRECISION, AND PROJECTION CONTROL

For row activations,

$$(xW^\top + b^\top)T = x(T^\top W)^\top + (T^\top b)^\top. \quad (17)$$

This algebra justifies temporary weight folding. The implementation computes products in float32 and casts back to the model dtype; exact invertibility statements concern real arithmetic. The insertion point is the native K/V linear output before rotary position operations and any subsequent key normalization. Grouped-query dimensions come from each selected linear module’s actual output dimension. The original weights are saved and restored before learning. Folding is an execution equivalence, not a separate claim of methodological novelty.

The projection design uses $P_r = U_{1:r}U_{1:r}^\top$, with r equal to half the module output dimension in the completed run. It preserves retained directions and nulls the remainder. This differs from T_r in both its spectrum and effective strength; the comparison does not constitute a distance-from-identity-matched experiment. At $\tau = 1$, SPECTRUM’s gains lie in $[1/2, 1]$. We do not claim to have measured the effect of random eigenvectors, isotropic attenuation, fixed geometry, or a τ sweep. Those controls are implementation options, not results in this paper.

B.4 WHY THE LOCAL RESULT IS NOT A SEMANTIC THEOREM

Distinct activations remain distinct after an invertible local map. Subsequent attention, nonlinear layers, vocabulary projection, stochastic decoding, and finite-data optimization can still map them to the same output class. A full-rank gain therefore cannot guarantee a particular value of C_k or D_b . Likewise, raw self-training has no unconditional improvement guarantee: in the idealized conditional setting where the sampling policy exactly equals the model distribution, the expected self-score gradient is zero at the generating parameters. The practical loop changes the sampling distribution, uses finite corpora and optimization, and in SPECTRUM uses fixed reference information. The empirical measurements are needed to establish the direction of change.

C ALGORITHM, INFORMATION FLOW, AND PROTOCOL

C.1 A COMPLETE ROUND

The following sequence states the exact separation of generator and learner.

1. Fix synthesis prompts $\mathcal{Q}$, reference anchor $\mathcal{A}$, held-out evaluation tasks, decoding settings, and the initial checkpoint θ_0 before the loop.
2. At round t , calculate reference-completion loss under θ_t , collect selected K/V-output gradients, and form $\bar{C}_{t,\ell}$ on the unchanged anchor.
3. Construct $T_{t,\ell}$ with the fixed strength τ and temporarily fold the operators into the current checkpoint’s native linear weights.
4. Sample S completions for each synthesis prompt. Keep every raw record. No execution outcome, reward, uniqueness judgment, or AST label determines membership or weight in $\mathcal{D}_t$.
5. Restore the original θ_t weights. Fit one LoRA adapter on $\mathcal{D}_t$, merge it into the base weights, and save θ_{t+1} .
6. Evaluate θ_{t+1} through common native inference, storing all outputs and verification labels. These labels do not affect the next training round. Repeat from step 2 using θ_{t+1} .

Table 4: **Completed training and evaluation settings.** The final 64-sample MBPP supplement is a separate sample pool; only 16-sample measurements exist for all historical rounds.

Component	Recorded setting
Backbone	Qwen/Qwen2.5-Coder-1.5B-Instruct; BF16, SDPA
Training seed / data split seed	43 / 42
Rounds / records per round	$5 / 291 \times 16 = 4, 656$
Calibration	50 fixed references; completion-only mean NLL; length limit 1,536
Geometry	Middle and last decoder layers; K/V outputs before RoPE; full second moment
SPECTRUM / projection control	$\tau = 1$ / rank fraction 0.5
LoRA	Rank 8, alpha 8, dropout 0.05; attention Q/K/V/O projections
SFT	1 epoch; learning rate 10^{-5} ; weight decay 0.01; warm-up ratio 0.03; gradient clipping 1.0
SFT batching / length	Microbatch 1, accumulation 16; maximum 1,536 tokens
SFT target mask	All nonpadding causal targets, including prompt targets
Common generation	Temperature 0.8, top- p 0.95, top- k 0
SSD training generation	Temperature 1.5, top- p 0.8, top- k 20
MBPP generation limits	512 new tokens, 1,024 prompt tokens
Transfer generation limits	1,024 new tokens, 4,096 prompt tokens
Native evaluation draws	MBPP every round: 16; final supplement: 64; transfer: 16
Uncertainty	2,000 task-bootstrap resamples; one main training seed

The anchor itself is never expanded with newly judged samples. It is reused as supervision at every calibration step, so “fixed” describes the information source rather than an initialization-only operation. In the completed study, there is no paired intervention-on generator diagnostic at every round; the native endpoints establish retained differences after learning, not the fraction of an immediate generator benefit transferred by SFT.

C.2 DATA AND PROMPTS

The MBPP preparation starts from the full official splits. Exact normalized prompt overlaps with held-out prompts are removed before partitioning the training pool; the recorded excluded training IDs are 602, 704, and 872. The resulting sizes are 291 synthesis, 50 calibration, 30 validation, and 500 evaluation tasks. A Python signature stub is extracted mechanically from the reference AST to specify the interface. Its implementation and evaluation assertions are not placed in the synthesis/evaluation prompt. The calibration partition supplies complete reference solutions.

The fixed MBPP prompt body is:

```
[task description]
Implement this Python function. Return only the complete
implementation.
[signature stub]
```

Brackets identify fields in a template, not missing manuscript results. The same template is used across methods; it does not request multiple strategies or select a strategy. Model-native chat formatting is applied by the tokenizer. Prompt and token provenance are stored in the experiment records.

HumanEval+ follows the recorded EvalPlus task interface and full base-plus tests, using release v0.1.10 with EvalPlus 0.3.1. APPS Intro is a fixed 200-task evaluation subset using the stored function-interface adapter. For stdin/stdout tasks, the prompt requests `solve(stdin: str)` `-> str`; the evaluator checks the returned output string against supplied cases using whitespace-token equality. This adapted interface, task subset, and evaluation budget are part of the result and differ from official benchmark leaderboard settings. No HumanEval+ or APPS references recalibrate the frozen MBPP students.

C.3 WHAT THE STUDENT OPTIMIZES

Let s denote the concatenated tokenized prompt and generated completion, and let $\mathcal{T}(s)$ contain its nonpadding causal targets. The recorded microbatch-one schedule accumulates example-level token-mean losses,

$$\ell_s(\theta) = -\frac{1}{|\mathcal{T}(s)|} \sum_{u \in \mathcal{T}(s)} \log \pi_\theta(s_u | s_{<u}), \quad \mathcal{L}_{\text{SFT}} = \frac{1}{|\mathcal{D}_t|} \sum_{s \in \mathcal{D}_t} \ell_s(\theta). \quad (18)$$

Gradient accumulation follows the recorded minibatch schedule. Including prompt targets and averaging each example by its own length are explicit choices, distinct from completion-only calibration and globally token-weighted corpus optimization. There are no sample-specific weights based on correctness, generated length, or source subspace. Truncation at the fixed context limit is a sequence-length constraint, not quality filtering. The completed training summaries record zero truncated training examples.

C.4 INFORMATION BUDGETS AND COMPARATORS

Plain and SSD do not use the reference anchor. SPECTRUM and the projection control use the same 50 references at each calibration step. Hence their direct comparison controls access to calibration information, whereas comparing either to Plain includes that difference in information use. No completed anchor-only SFT control isolates this factor. SSD here adapts the published decoding idea to the small-model MBPP budget; it does not reproduce the original larger-scale results. UA-RL, which uses rewards for rare strategies, is discussed as related work but is not a completed numerical comparator in this archive.

D COMPLETE MEASURED RESULTS

D.1 LONGITUDINAL MBPP RESULTS AT 16 SAMPLES

Table 5 contains all recorded rounds. The initial SSD row is an independently sampled evaluation of the same starting model, retained separately from the original trajectory. Small differences in initialization estimates are sampling/protocol-stream differences, not different learned checkpoints. No 64-sample intermediate-round results are inferred from these rows.

D.2 FINAL PROJECTION COMPARISON AND CORRECT-COUNT MATCHING

Sparse D_{64} estimates are available on few fully correct tasks but do not form a primary claim: the paired SPECTRUM–Plain population is only 29 tasks. At the recorded 16-sample round-five endpoint, the SPECTRUM-minus-projection difference in exact-program D_4 is 0.312 with interval $[0.253, 0.377]$, and the AST difference is 0.293 $[0.233, 0.353]$. The coarser control-flow difference is 0.042 $[-0.011, 0.095]$. This is why we describe the outcome as implementation breadth rather than proven algorithm diversity.

D.3 TRANSFER CONTRASTS AND THE PROJECTION CONTROL

The projection control’s HumanEval+ scores are pass@1 52.74%, pass@16 84.76%, $C_{16} = 5.921$, and $D_4 = 3.473$ on 115 eligible tasks. Its APPS Intro scores are 14.88%, 34.50%, 1.880, and 3.685 on 47 eligible tasks. The SPECTRUM-minus-projection paired D_4 effects are 0.039 $[-0.021, 0.102]$ on 109 HumanEval+ tasks and 0.067 $[-0.003, 0.142]$ on 42 APPS tasks. Thus the transfer evidence does not establish a uniform advantage over this control.

Table 5: **Every completed native-student round on MBPP, 16 draws per task.** Full 500-task means for pass@1 and C_{16} ; marginal D_4 means with eligibility. SSD is the separate continuation.

Method	Round	pass@1 (%)	C_{16}	D_4	Eligible
Initial	0	37.92	4.016	3.310	262
Plain	1	38.42	3.808	3.173	261
Plain	2	39.15	3.634	3.080	269
Plain	3	39.65	3.372	2.895	257
Plain	4	40.79	3.190	2.803	270
Plain	5	41.30	3.090	2.747	268
SPECTRUM	1	37.84	3.848	3.222	261
SPECTRUM	2	38.36	3.768	3.159	259
SPECTRUM	3	39.05	3.662	3.063	262
SPECTRUM	4	39.77	3.588	2.991	263
SPECTRUM	5	40.17	3.606	2.997	263
Projection control	1	37.92	3.692	3.144	257
Projection control	2	39.20	3.592	3.059	267
Projection control	3	39.40	3.314	2.889	256
Projection control	4	41.09	3.162	2.774	266
Projection control	5	41.35	3.066	2.690	262
SSD initial	0	37.81	3.992	3.311	260
SSD recipe	1	37.80	3.846	3.232	260
SSD recipe	2	36.81	3.710	3.200	250
SSD recipe	3	36.60	3.724	3.196	256
SSD recipe	4	37.09	3.752	3.181	257
SSD recipe	5	35.96	3.710	3.268	255

Table 6: **Final 64-sample projection ablation.** Same 500 tasks, fixed reference access, and five-round training schedule. Marginal D_4 means are descriptive; eligible tasks are 321 and 323.

Generator	pass@1 (%)	pass@64 (%)	C_{64}	Retention (%)	D_4
Subspace projection	41.77	70.20	8.388	65.5	2.810
SPECTRUM	40.33	72.60	11.510	89.9	3.143

Table 7: **Paired final MBPP richness contrasts, SPECTRUM minus Plain.** Common eligible tasks are selected separately for each correct-draw budget. Intervals are the archived 95% task-bootstrap intervals.

Endpoint	Paired difference	95% interval	Tasks
C_{64}	3.004	[2.568, 3.494]	500
D_4	0.295	[0.249, 0.343]	316
D_8	0.780	[0.665, 0.889]	296
D_{16}	1.816	[1.582, 2.055]	254

Table 8: **Paired transfer effects of SPECTRUM relative to Plain.** Differences use all tasks for pass@1 and C_{16} , and common eligible tasks for D_4 . Intervals are derived from existing task records with 2,000 resamples; no new generations or verification were performed.

Benchmark	Endpoint	Difference	95% interval	Tasks
HumanEval+	pass@1 (pp)	-1.524	[-3.088, 0.076]	164
	C_{16}	0.152	[-0.128, 0.439]	164
	D_4	0.080	[0.021, 0.142]	111
APPS Intro	pass@1 (pp)	-1.219	[-1.937, -0.530]	200
	C_{16}	-0.085	[-0.210, 0.055]	200
	D_4	0.104	[0.007, 0.205]	42

E ACTUAL EQUAL-CORRECT-COUNT EXAMPLES

The compact archive retains correctness and program fingerprints but removes program text. We therefore show actual class-frequency records rather than reconstructing purported code samples. For each sign of the SPECTRUM-minus-Plain observed AST-count difference, choose the numerically lowest HumanEval task ID with equal correct counts of at least four. This rule yields HumanEval/3 for a positive example and HumanEval/2 for a negative example (Figure 4). Selection conditions on the sign and is illustrative, not an estimate of prevalence.

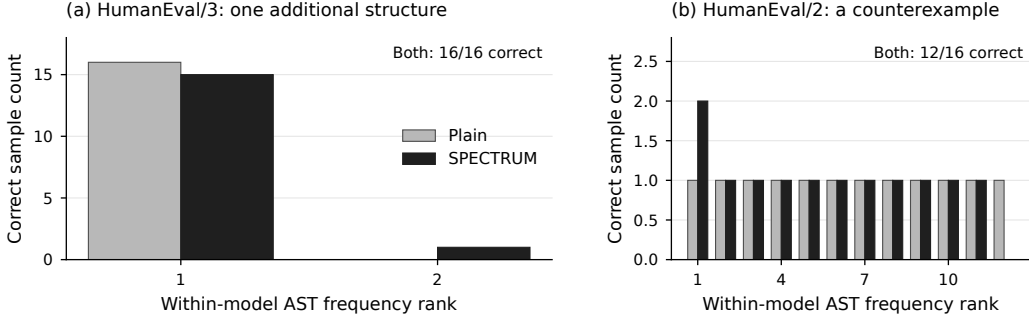


Figure 4: **The same number of correct outputs can conceal different structures.** Actual saved round-five HumanEval+ samples. (a) Both methods pass all 16 draws on HumanEval/3. Plain has one AST class with count 16; SPECTRUM has counts 15 and 1, giving $\hat{D}_4 = 1$ and 1.25. (b) Both pass 12 of 16 draws on HumanEval/2; Plain has twelve singleton classes, while SPECTRUM has one doubleton and ten singletons. Their $\hat{D}_4$ values are 4 and 3.909. Classes are ranked separately within each method, so horizontal position does not assert matching class identity. Both cases follow the stated lowest-ID rule.

These examples instantiate the measurement distinction without supplying semantic algorithm labels. At equal observed n and m , every finite-sample pass@ k estimator agrees between methods, while the correct-class occupancy can differ in either direction. Equal estimated pass curves are not proof that the underlying true success probabilities are identical.

F RESOURCES AND REALIZED SAMPLE LENGTHS

All methods use the same 23,280 generated training records over five rounds. Table 9 records their realized lengths. SPECTRUM generates 9.9% more completion tokens than Plain; equal record count is therefore not equal token count or FLOPs. Matching the number of correct evaluation samples does not control training length or compute. It answers a distributional question about the saved students.

Table 9: **Training-generation resources across five rounds.** Counts are summed from saved round budgets; mean length is generated tokens divided by records. Length-capped records remain in training. SFT token counts include prompt targets under the recorded loss mask.

Method	Records	Generated tokens	Mean length	Capped	SFT tokens
Plain	23,280	4,859,928	208.76	617	6,373,288
SSD recipe	23,280	5,812,400	249.67	1,498	7,325,760
SPECTRUM	23,280	5,342,368	229.48	833	6,855,728
Projection control	23,280	4,727,089	203.05	619	6,240,449

For a selected module of output dimension d and M gradient positions, dense second-moment accumulation costs $O(Md^2)$ and eigendecomposition costs $O(d^3)$, excluding the reference forward/backward passes. Dense folding of a $d \times p$ weight costs $O(d^2p)$. These operations are performed during calibration/construction, not at each generation token. The final student has the original dense architecture.

Archived runtime records cover different hardware and stages; we make no matched wall-clock or GPU-memory efficiency claim from their combination.

G POSITIONING AND EVIDENCE BOUNDARIES

Sequence-level distillation versus token-feedback OPSD. The common phrase “learning from one’s own outputs” covers multiple objectives. In sequence-level self-distillation, sampled tokens become cross-entropy targets. In demonstration-conditioned on-policy distillation, student rollouts receive dense token feedback from a teacher with privileged context. The present implementation is the former. Its sampling policy is derived from the current model, but temporarily changes selected weights; “on-policy” should not erase this generator–student distinction. We use Looped Self-Distillation as a precise outer-process description.

Relation to earlier iterative methods. Iterative self-distillation is not introduced for the first time here. The context/rationale study of Rao et al. (2023) includes iterative self-imitation and examines quality and diversity. SCoder progressively improves a data synthesizer using multiple checkpoints and scoring/selection, and discusses declining diversity (Zhang et al., 2025). CRISP demonstrates recurring teacher updates without correctness-based sample selection (Sang et al., 2026). These works prevent defining novelty solely through recurrence or the absence of a verifier filter. Our narrower contribution joins conditional correct-implementation retention, a fixed-anchor internal generation intervention, and repeated measurements of the native learned students.

Relation to demonstration-conditioned diversity loss. Nicolicioiu et al. (2026) analyze self-distillation with sampled correct demonstrations, including feedback bias and diversity changes over training. Their mechanism should not be described as having no temporal iteration. Our five outer rounds repeatedly generate a complete raw corpus and fit a new native student without a correct-demonstration teacher. The occurrence of contraction in this setting shows that rollout filtering or that particular teacher-feedback construction is not necessary for the measured phenomenon. It does not identify a unique causal source of contraction, nor establish that all self-distillation schemes must contract.

Completed and excluded evidence. The manuscript uses the full MBPP seed-43 five-round archive, its separate final 64-sample supplement, the SSD seed-43 continuation, and saved HumanEval+/APPS transfer records. Earlier 64-task pilot results are not pooled with the full benchmark. Four-task smoke runs are software checks, not research evidence. Configured multi-seed confirmation, larger models, additional architectures, further datasets, UA-RL, and unexecuted geometry controls are excluded from numerical tables. Their existence as runnable code does not establish measured performance. All figures and additional intervals in this revision are analyses of already recorded results.